Iryna Shevchenko

Senior Software Engineer

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Summary

Senior Software Engineer with 10 years of experience designing and delivering scalable, high-performance applications across AI-driven chatbots, healthcare, and cloud-native ecosystems.

Expert in Python, Django, Flask, FastAPI, React, Angular, and emerging technologies like LangChain and LangGraph. Proven ability to fine-tune LLMs (Large Language Models), optimize machine learning pipelines, and deploy cloud-based solutions using AWS, GCP, and Kubernetes.

Skilled in microservices architecture, RESTful APIs, and advanced data processing pipelines. Adept at improving system efficiency, reducing production incidents, and delivering robust, user-centric solutions. A collaborative leader in agile environments, driving innovation and customer satisfaction.

Skills

Programming Languages: Python, JavaScript (ES6+), TypeScript, HTML5, CSS3, SQL, Java

Frameworks/Libraries: Django, Flask, FastAPI, React, Angular, unittest, pytest, Mock, SQLAlchemy, LangChain, LlamaIndex, CrewAI, AutoGen, Dialogflow, Flowise, Ollama, LiteLLM, Transformers, Bitsandbytes, vLLM, TensorRT, LangSmith, Helicone

Databases: PostgreSQL, MongoDB, Redis, MySQL, Pinecone, Weaviate

Cloud Platforms: Amazon Web Services (AWS) (EC2, EKS, Lambda, SNS, S3), Google Cloud Platform (GCP) (Compute Engine, GKE, Pub/Sub, Cloud Functions), Microsoft Azure (AKS, Functions, Blob Storage, Service Bus), lambdalabs, Runpod

Tools/Technologies: RESTful APIs, GraphQL, Microservices, Docker, Kubernetes, CI/CD, Jenkins, GitHub Actions, Jira, Git, Apache Kafka, RabbitMQ, Elasticsearch, Logstash, Kibana (ELK Stack), OAuth 2.0, JWT, TDD, Asynchronous Programming

Experience

Senior Software Engineer

Luxoft (New York) **Oct 2021 – Present**

- Spearheaded the development of a cutting-edge AI-powered sales assistant, leveraging LangChain, LlamaIndex, and LLMs (LLaMA2) to enable natural language querying of SQL databases, ensuring hallucination-free responses and seamless customer interactions.
- Architected and deployed cloud-native applications on AWS, utilizing EC2 for scalable compute resources, S3 for data storage, RDS for relational databases, and Lambda for serverless computing, ensuring efficient, cost-effective infrastructure management.
- Implemented robust CI/CD pipelines using AWS CodePipeline and CodeDeploy, streamlining the deployment process and enabling rapid feature releases with minimal downtime.
- Optimized cloud costs by leveraging AWS auto-scaling groups, spot instances, and reserved instances, reducing infrastructure expenses by 25% while maintaining high performance.

- Fine-tuned LLaMA2 models using feedback loops via LangSmith, significantly improving model accuracy and end-user satisfaction; contributed to a sustained boost in CSAT scores to 4.6/5.
- Architected and deployed a multi-turn chat engine using Pinecone for vector search, LangGraph for multi-step interaction logic, and FastAPI, powering scalable LLM-driven experiences across internal platforms.
- Optimized the Bark Text-to-Speech (TTS) model by integrating TensorRT for GPU acceleration and deploying on Google Kubernetes Engine (GKE), achieving a 350% performance gain while handling over 30,000 daily requests.
- Designed and maintained a high-throughput Kafka-based data pipeline to ingest and process millions of real-time events per second, improving analytics reliability and data availability by 40% for downstream systems.
- Developed and deployed a fraud detection ML model using scikit-learn and TensorFlow, integrating it into transactional workflows and reducing fraudulent activity by 25% across monitored systems.
- Built a real-time monitoring dashboard using FastAPI, Flask, and React, visualizing system KPIs and infrastructure health, which enabled proactive maintenance and reduced unplanned downtime by 15%.
- Led the modernization of a monolithic legacy platform, redesigning it into modular, Dockerized microservices, deployed via CI/CD pipelines into a containerized cloud environment, enhancing scalability and fault isolation.
- Ensured robust data protection and compliance through implementation of OAuth 2.0, JWT authentication, and role-based access control for systems handling sensitive financial and PII data.
- Tuned PostgreSQL performance through query optimization, indexing strategies, and partitioning, decreasing critical query response times by 30% across high-load endpoints.
- Introduced comprehensive automated testing using pytest, unittest, and Mock, achieving over 90% code coverage and reducing post-release bugs by 30%.
- Mentored junior engineers in Python, cloud-native development, and LLMOps best practices, championed use of LangSmith for prompt debugging and evaluation, and fostered a collaborative, quality-first engineering culture.

Senior Software Developer

IncWorx Consulting (Schaumburg, IL) **Aug 2017 – Sep 2021**

- Designed, developed, and maintained HIPAA-compliant microservices in Python (Flask, FastAPI) to support member eligibility, claims processing, and provider search, reducing latency by 40% across core services.
- Integrated AWS S3 with machine learning models, enabling seamless storage and retrieval of large datasets for model training and inference, while ensuring optimal performance and security through IAM roles and bucket policies.
- Led migration of legacy monolith systems to modular Python microservices architecture, improving maintainability and reducing deployment times by 60%.
- Engineered robust REST APIs and background workers for high-throughput data ingestion pipelines, adhering to strict PHI data handling protocols.
- Optimized complex queries in PostgreSQL using SQLAlchemy, resulting in 60% faster read operations for high-volume reporting dashboards.
- Integrated Redis as a distributed caching layer for claims and benefits endpoints, reducing database load and improving average response times from 1.4s to under 600ms.
- Built real-time event processing and alerting systems with Apache Kafka, enhancing incident detection speeds by 35% and improving system observability in critical care workflows.
- Implemented automated CI/CD pipelines with GitLab and AWS, deploying containerized services via ECS and Lambda, with blue/green deployment strategies to ensure zero-downtime releases.
- Developed and enforced unit, integration, and contract testing strategies using unittest, pytest, and tox, reaching 90%+ test coverage and cutting production incidents by 60%.
- Partnered with security and compliance teams to support SOC 2 and HIPAA audits, ensuring all services followed best practices for encryption, access control, and logging.

Full Stack Developer

Amazon (Seattle, WA) **May 2014 – Jul 2017**

- Contributed to the development of enterprise-grade internal tools and customer-facing portals used across multiple business units at Oracle, working across both Python (Flask) and JavaScript (initially jQuery, later React) stacks.
- Assisted in modernizing legacy systems into microservice-style applications, helping split out backend logic into modular services with clearly defined RESTful API contracts.
- Utilized AWS S3 to store and manage large datasets for internal dashboards, integrating with both Oracle SQL and PostgreSQL databases to enhance customer data analytics and visualization performance.
- Implemented asynchronous processing for large-scale background jobs by integrating AWS SQS and Lambda functions into RabbitMQ queues, improving job execution efficiency and reducing latency in system response
- Enhanced deployment workflows by utilizing AWS CodePipeline and CodeDeploy for automated application delivery, reducing manual intervention and speeding up release cycles across multiple development stages.
- Built and maintained internal dashboards to visualize customer data and system metrics, using D3.js, React, and Bootstrap to create intuitive interfaces for non-technical stakeholders.
- Wrote backend service logic to process transactional data, connect with internal Oracle databases (Oracle SQL, PostgreSQL), and expose analytics endpoints to front-end apps.
- Supported integration of RabbitMQ queues to decouple services and enable asynchronous processing for large, scheduled background jobs (e.g., report generation, user activity tracking).
- Created internal documentation, quick-start scripts, and test data to help onboard new team members and support QA with integration and system testing.
- Regularly participated in code reviews and sprint planning meetings, offering feedback and raising concerns around performance bottlenecks or service dependencies.
- Contributed bug fixes and performance improvements to legacy Oracle ERP tools, including optimizing long-running stored procedures and indexing frequently accessed database tables.
- Helped improve deployment confidence by writing and maintaining unit tests and lightweight integration tests, using pytest and mock for backend services.
- Collaborated closely with DevOps engineers to understand deployment pipelines, write deployment configs, and troubleshoot staging/production issues across Linux environments.

Education

Bachelor of Computer Science | **University of North Texas at Dallas** | Sep 2010 – Apr 2014